MIL-W-36241C
17 September 1976
SUPERSEDING
MIL-W-36241B (DSA-DM)
11 December 1973

MILITARY SPECIFICATION

WASHCLOTH, HOSPITAL PATIENT, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type and size of a disposable, hospital patient washcloth.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

STANDARDS

FEDERAL

FED-TEST METHOD STD-191 - Textile Test Methods.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7210

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS (ASTM)
D1117 Standard Methods of Testing Nonwoven Fabrics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)
93 Abrasion Resistance of Fabrics: Accelerotor Method

(Application for copies should be addressed to The American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Fabric. The nonwoven fabric shall be fabricated from any of the following compositions at the manufacturer's option. The color of the washcloths shall be optional with the manufacturer. The nonwoven fabric shall be soft, have good hand (feel) and shall be non-irritating and non-toxic, when inspected as specified in 4.3.1.3. The fabric shall conform to the requirements of 3.1 and table I when inspected as specified in 4.3.1 through 4.3.1.2 and 4.3.1.4.

a. The nonwoven fabric shall consist of 100 percent, 1-1/2 denier, 1-9/16 inch average staple length viscose rayon discontinuously printed and bonded with a cellulosic type binder into a foraminous structure.

b. The nonwoven fabric shall consist of 100 percent, 1-1/2 denier, 1-9/16 inch average staple length viscose rayon discontinuously bonded with a laytex resin adhesive into a geometrical or irregular pattern such as a cross hatch or floral print design.

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c. The nonwoven fabric shall consist of cellulosic or synthetic fiber or a combination thereof, wet or dry formed, reinforced with cotton or synthetic scrim or combination thereof or fiber, and resin bonded or bonded with an acrylic type of adhesive. The fabric may contain stripes, checks or a floral print design on one or both sides.

TABLE I. Nonwoven fabric properties

Characteristics	Requirements		
	Composition "a"	Composition "b"	Composition "c"
Weight per square yards, minimum:			
Ounces	1.4	1.4	1.7
Grains	610	610	750
Breaking strength, pounds minimum:	Dry / Wet	Dry / Wet	Dry / Wet
Machine direction	16 1/ / 7.5 1/	25 1/ / 10 1/	4.0 1/ / 13.0 2/
Cross direction	2.5 1/ / 1.0 1/	2.25 1/ / 0.5 1/	3.5 1/ / 6.0 2/
			3.0 1/ / 6.0 2/
			2.8 1/ / 4.0 2/
Water absorption, percent minimum	400	700	300
Weight loss, percent maximum	10.0	10.0	10.0
Number of holes per square inch	120 ± 15	----	----

1/ Applicable when the breaking strength is determined by the Cut Strip Method, Method 5102 of FED-TEST METHOD STD - 191.

2/ Applicable when the breaking strength is determined by the Grab Method, Method 5100 of FED-TEST METHOD STD - 191.

3.2 Performance inspection.

3.2.1 Washing characteristics. When inspected as specified in 4.3.3, the washcloth shall not disintegrate, shred or ball-up; shall not become slippery and difficult to hold; shall not scratch nor appear dirty or oily before or after wetting and soaping; and shall have no objectionable odor before or after wetting and soaping. The washcloth upon drying shall develop no objectionable odor; shall retain its original shape and appearance; and shall be capable of reuse without any of the aforementioned occurring.

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3.2.2 Resistance to sterilization. When steam sterilized as specified in 4.3.4, the washcloth shall not disintegrate, shred, shrival or ball up, appear dirty or oily, become stiff or slippery or develop an objectionable odor. After drying and conditioning, the washcloth shall then be inspected and shall conform to the requirements for washing characteristics, 3.2.1.

3.3 Dimensions. When inspected as specified in 4.3.5, the size of the washcloth shall be not less than 136 square inches in area and not less than 10-1/2 inches in one direction.

3.4 Workmanship. The washcloths shall be free from any unspecified hole, tear or cut, acids, chemicals, detergents or any other ingredient that shall impair the bonding, serviceability or use. The washcloths shall not contain any spot of oil or grease, fiber clump, insect, smudge, dirt, sharp object, bonding material strike through, congealed mass of the bonding material, or spot or stain resulting from excess bonding material and any other defect which detracts from its appearance or may impair its serviceability. The washcloths shall conform to the quality and grade of product established by this specification. The occurence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material and component inspection. In accordance with 4.1, the materials and components shall be inspected and tested in accordance with all the requirements of the referenced specifications and standards, unless otherwise excluded, amended, modified, or qualified in this specification or purchase order.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contactor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.1.3 Inspection. Inspection, as used in this specification, is defined as both examination (an investigation without the use of special laboratory apparatus or procedures such as visual) and testing (a determination by technical means of the physical and chemical properties) of the item.

4.2 Quality conformance inspection. The sampling and inspection shall conform to MIL-STD-105, except where otherwise indicated hereinafter. The quality conformance inspection shall consist of the tests and examinations specified in table II.

4.2.1 Sampling.

4.2.1.1 Inspection lot.

4.2.1.1.1 Fabric. An inspection lot size shall be expressed in units of one linear yard and shall consist of all the fabric received by the washcloth manufacturer at one time. The sample unit for inspection shall be one linear yard of the fabric.

4.2.1.1.2 Washcloths. An inspection lot size shall be expressed in units of one washcloth made essentially under the same conditions and from the same materials and components. The sample unit for inspection shall be one washcloth.

4.2.1.2 Sampling. Sampling for inspection shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for inspection

Item	Inspection	Requirement	Test	Inspection level	AQL (defects per hundred units)
Fabric	Weight	3.1	4.3.1	1/	1/
	Breaking strength				
	Dry	"	"	"	"
	Wet	"	"	"	"
	Water absorption	"	"	"	"
	Holes per square inches	"	"	"	"
	Weight loss	"	4.3.1.2	"	"
	Toxicity	"	4.3.1.3	"	"
	Fiber Identification	"	4.3.1.4	"	"
	Staple Length	"	"	"	"
	Type of Binder	"	"	"	"

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TABLE II. Sampling for inspection (continued)

Item	Inspection	Requirement	Tests	Inspection level	AQL (defects per hundred units)
Wash-Cloth	Examination	3.3 through 3.5	4.3.2.1	S-3	4.0
	Washing Characteristics	3.2.1	4.3.3	S-1	1.5
	Resistance to Sterilization	3.2.2	4.3.4	S-1	1.5
	Dimension	3.2.3	4.3.5	S-3	4.0

1/ The sampling for the testing of the fabric shall be in accordance with 4.2.1.1.1 and as follows:

Lot size (yards)	Sample size
800 or less	2
801 and up to and including 22,000	3
22,001 and over	5

The samples shall be taken from the beginning of each roll. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified in table II. Each sample unit taken for inspection shall be used to determine all the characteristics of the fabric specified in table II.

4.3 Inspection methods.

4.3.1 Fabric. The fabric characteristics shall be determined in accordance with the inspection methods of FED-TEST METHOD STD-191 specified in table III and 4.3.1.1 through 4.3.1.4.

TABLE III. Fabric properties

Characteristics	FED-TEST METHOD STD - 191 Method
Weight	5041
Breaking strength:	
As received	5100 or 5102 1/
Wet 2/	5100 or 5102 1/
Number of holes	5050

1/ Each specimen shall be one inch wide. The breaking strength of the composition b fabric shall be determined by using 3 ply specimens.

2/ Each specimen shall be immersed for 10 seconds in distilled water at a temperature of 70° to 80° Fahrenheit (F.) (21.1° to 26.7° Centigrade (C.)) in

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such a manner that the water shall be all about each separate specimen. Upon completion of the 10 second immersion, each specimen shall be hung up and allowed to drain for 15 seconds. Upon completion of the 15 second draining, each specimen shall immediately be inspected for wet breaking strength.

4.3.1.1 Water absorption. The water absorption shall be determined in accordance with ASTM D1117 or as specified in 4.3.1.1.1 through 4.3.1.1.1.4.

4.3.1.1.1 Principle of the method. A weighed sample of the fabric is dropped onto the surface of water in a container. The increase in weight divided by the original weight of the sample times 100 is the percent of water absorbed.

4.3.1.1.1.1 Conditions of the samples. The samples of the fabric shall be conditioned for a minimum of two hours under standard atmospheric conditions as specified in FED-TEST METHOD STD-191, prior to determining the water absorption.

4.3.1.1.1.2 Apparatus. The apparatus to be used shall consist of the following:

a. A cylindrical basket 8 centimeters long and 5 centimeters in diameter weighing 3 grams fabricated as follows: The spaces between the meshes shall be approximately 2 centimeters. The top and bottom circles of the basket shall be fabricated from two pieces of copper wire, approximately 0.4 millimeters in diameter and 33 centimeters long. The top and bottom circles shall be doubly wound. The circles shall be 5 centimeters in diameter and shall be spaced two centimeters apart. The overall length of 8 centimeters is of more importance than the two centimeter spacing. The center uprights shall be fabricated from three pieces of B and S copper wire approximately 0.4 millimeters in diameter and 16.7 centimeters long. The uprights shall be placed approximately 19 millimeters apart on the circumference of the circles. The uprights shall be basket woven through the circles. Before winding excess wire in the uprights around the end circles, the basket shall be weighed. If necessary snip off portions of the uprights until the basket weighs 3 $\pm$ 0.1 grams. Then complete the basket by winding any excess wire around the end circles.

b. The container to hold the water shall be a minimum of 5 inches in diameter and 4-1/2 inches in height.

c. One stop watch calibrated to 1/5 of a second.

d. One tared watch glass 5 inches in diameter.

e. One balance with the capability of weighing to 0.1 of a gram.

f. One pair of tongs approximately 12 inches long.

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4.3.1.1.1.3 Sample preparation. Cut a 5 ± 0.1 gram sample into strips approximately 3 by 10 inches.

4.3.1.1.1.4 Procedure. Combine each of the strips and rake into a stack. Loosely roll the stack to form a cylinder 3 inches long with a diameter to fit the basket. Place the sample in the basket and rotate so that the sample loosens sufficiently to fit snugly in the basket. Fill the container with distilled water to depth of at least 10 centimeters. The temperature of the water shall be maintained at 77° ± 1.8°F. (25° ± 1° C.). Hold the basket on its side 1/2 inch above the surface of the water and drop into the water. Ten seconds after the basket and sample have been submerged, remove the basket and sample immediately from the water. Hold the basket and sample in a horizontal position and allow the water to drain for 10 seconds. Upon completion of the 10 second draining, place the basket and sample on the tared watch glass and record the weight. The percent of water absorption shall be calculated as follows:

$$\text{PERCENT WATER ABSORPTION} = \frac{Wt - (Wb + Ws + Wg)}{Ws} \times 100$$

Wt = final total weight
Wb = weight of basket
Wg = weight of watch glass
Ws = weight of sample

4.3.1.2 Weight loss. The weight loss after 20 seconds at 3,000 revolutions per minute shall be determined in accordance with AATCC Test Method 93.

4.3.1.3 Toxicity. The toxicity and irritability of the fabric shall be determined by a standard method approved by the contracting officer.

4.3.1.4 Fiber identification, staple length and binder. The fiber identification, staple length and binder shall be determined by any nationally recognized standard method.

4.3.2 Visual examination.

4.3.2.1 Washcloths. Each washcloth selected as a sample unit shall be thoroughly examined visually to determine conformance to this specification. The list of defects, table IV, shall be used to enumerate the defects found.

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TABLE IV. List of defects for the visual examination of the washcloths

Defects
Any cut, unspecified hole or tear. Any spot, stain or clump of fiber greater than 1/2 inch in diameter. Any mend or patch, sharp or foreign object, or insect. Any edge that is ragged, on an angle or not straight. Any defect that affects the serviceability or appearance of the washcloth not herein listed.

4.3.2.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE V. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.2.2.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.3.3 Washing characteristics. Place two quarts of water having a temperature of 105° to 115°F. (40.6° to 46.1°C.) in a basin or pan. Thoroughly wet the washcloth to determine if the odor is objectionable. Thoroughly wet the washcloth in the warm water and squeeze to remove the excess water. Smell the washcloth again. Rub the open damp washcloth briskly ten times with a bar of Lifebouy,

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Dial, Ivory or any other similar commercial soap and observe if the washcloth soaps easily, disintegrates, shreds, balls up, becomes slippery and difficult to hold, scratches or appears dirty, oily and whether an offensive odor develops (see 3.2.1). Then wash both hands and observe as aforementioned. Repeat the wetting, soaping, washing and observations again. Upon completion of the aforementioned, the washcloth shall be dried and conditioned for a minimum of two hours under standard atmoshperic conditions as specified in FED-TEST METHOD STD - 191 and then repeat the washing characteristics inspection.

4.3.4 Resistance to sterilization. The washcloth shall be exposed to steam sterilization for a minimum of 20 minutes after the temperature has reached 250°F. The washcloths shall be exposed in such a manner that shall permit the steam to circulate completely around each washcloth. Upon completion of the exposure time, the washcloths shall be removed from the autoclave and observed to determine if the the washcloth has shredded, disintegrated or shrivelled or balled up; is stiff or slippery; appears to be dirty or oily; or an offensive odor has developed. Upon completion of the observations, the washcloths shall be dried and conditioned for a minimum of two hours under standard atmospheric conditions as specified in FED-TEST METHOD STD-191 and then shall be subjected to the washing characteristics inspection, 4.3.3.

4.3.5 Dimensions. The washcloth shall be laid flat without tension on a smooth flat surface. The measurements shall be made with a device capable of measuring to nearest 1/16 inch.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing one thousand (1000) washcloths, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation. One thousand thoroughly clean washcloths shall be individually folded into a square or rectangle. Fifty or one hundred folded washcloths shall be wrapped or bagged in polyethylene and the wrap or bag sealed.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Level A. One thousand wrapped or bagged washcloths shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. One thousand wrapped or bagged washcloths shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Polyethylene wrapped or bagged washcloths. Each polyethylene wrap or bag shall be marked to identify contents and quantity contained. Marking may be printed on the wrap or bag or on a label placed within the wrap or bag in such a manner so as to permit ready identification.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The disposable, hospital patient washcloth is intended for use by patients confined to military hospitals.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Catalog information. The washcloth covered by this specification is listed in the Federal Supply Catalog under the following stock number.

National Stock Number	Item Identification
7210-01-013-2824	Washcloth, Hospital Patient, Disposable 1000s.

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6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-W-36241C WASHCLOTH, HOSPITAL PATIENT, DISPOSABLE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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